Simplifying complexity

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Large communities and random interactions

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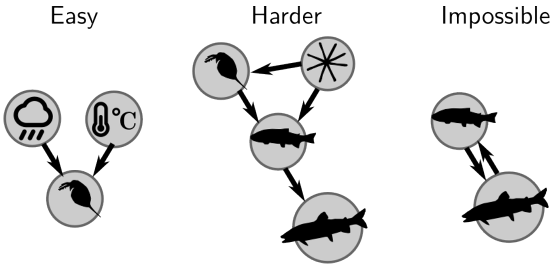
19/03/2026

Preamble 1: Why this lecture

Stats work for simple causal arrows, or at worst, for directed graph



- When too many factors, inference of every link gets difficult
- When mutual causation, we cannot use X to predict Y or conversely

  (we can use past X & Y to predict future X & Y = dynamics, but also hard to fit)

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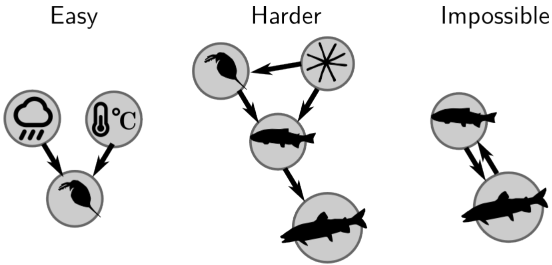

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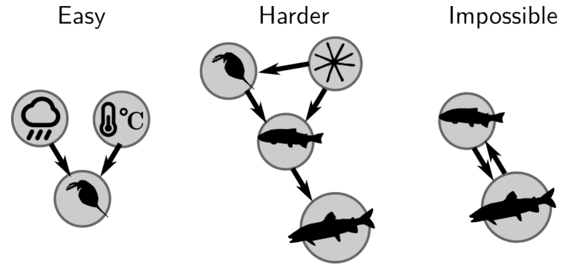

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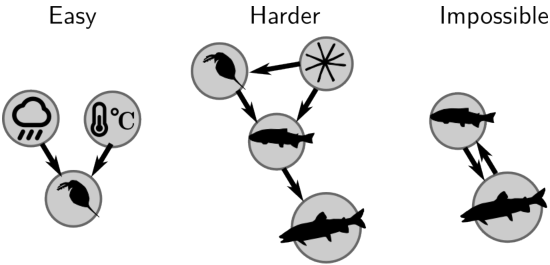

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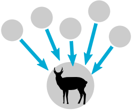
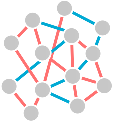
Two questions: for my data, do I believe

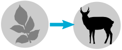
- I can only focus on a few variables & factors that matter, or not?
- I can neatly separate what I study into causes and consequences (independent & dependent variables), or not?



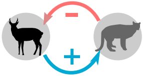


Many things matter

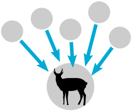
Few things matter

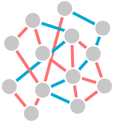


One-way causality



Feedbacks





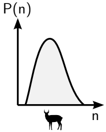
forget who does what,
count what happens

Many things matter

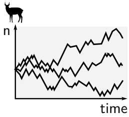
PCA,
network
metrics,
...

Few things matter

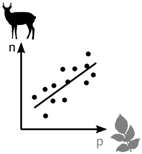
identify who does what



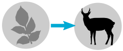
histograms



random walks



regressions



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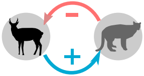

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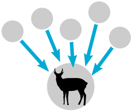

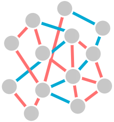

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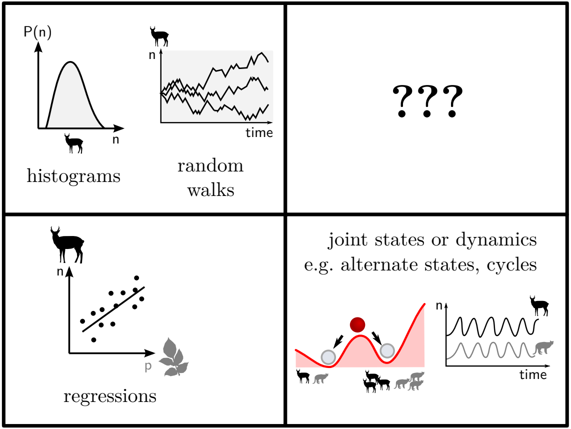
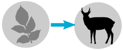
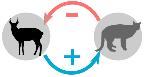




One-way causality

more stats ⟷ more dyn



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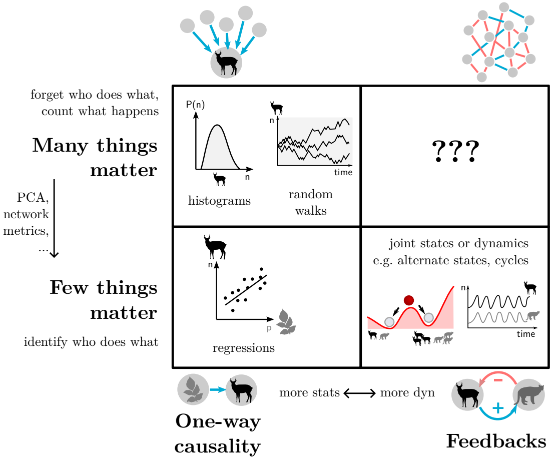

Is top-right corner hopelessly complex and beyond understanding?

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Preamble 2: an illustration in simulated data

Model

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e.g. replicators, multi-strain SIR, Agent-Based Models...

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... so I will just use Lotka-Volterra

Model

$$\frac{dN_i}{dt} = r_i N_i (1 - a_{ii} N_i - \sum_{j \neq i}^{S} a_{ij} N_j) \tag{1}$$

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$$r = (?, ?, ?....) \tag{2}$$

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we need many parameters:

- growth rates r_i (S numbers, 1 per species)

$$r = (?, ?, ?....) \tag{2}$$

- even harder, interactions a_{ij} (S^2 numbers, S per species)

$$a = \begin{pmatrix} ? & ? & ... \\ ? & & \\ ... & & \end{pmatrix} \tag{3}$$

Model

Practical modeller's strategy: don't know r_i and a_{ij} in your system? take them at random!

$$a = \begin{pmatrix} 0.29 & 0.54 & 0.53 & 0.02 & ... \\ 0.57 & 0.86 & 0.90 & 0.81 & ... \\ ... & & & & \end{pmatrix} \tag{4}$$

Model

Practical modeller's strategy: don't know r_i and a_{ij} in your system? take them at random!

$$a = \begin{pmatrix} 0.29 & 0.54 & 0.53 & 0.02 & \dots \\ 0.57 & 0.86 & 0.90 & 0.81 & \dots \\ \dots & & & & \end{pmatrix} \tag{4}$$

Problem: results could depend on many details of the matrix, e.g. each value, or how we drew the random numbers (normal, uniform, etc.)

Predictions

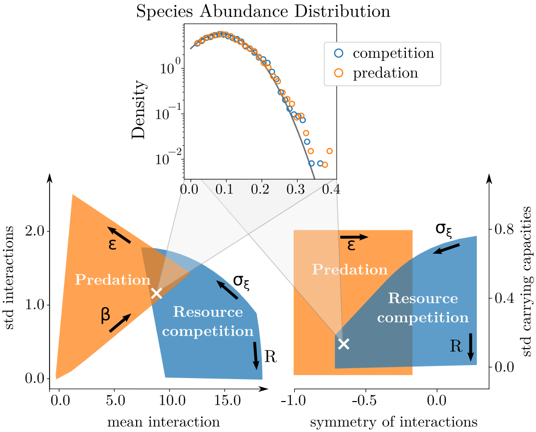
In fact, if you simulate again and again, many results only depend on a few global statistics of the community:

- number of species S
- mean of interaction strength $\langle a_{ij} \rangle$ ($= \mathrm{E}[a_{ij}]$ if you prefer)
- standard deviation $\mathrm{std}(a_{ij})$
- and symmetry $\mathrm{corr}(a_{ij}, a_{ji})$

(assuming many species and not too crazy random distributions)

Predictions

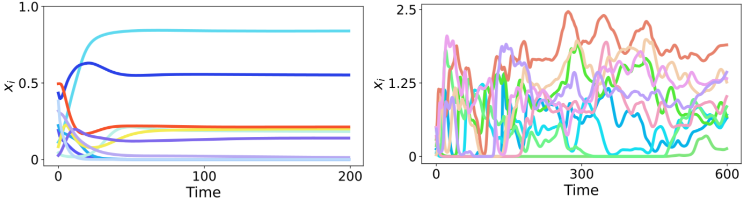
In particular, nature of interactions (competitive, trophic, parasitism...) is *irrelevant*!



e.g. two models, one with predation, one with competition, give same results (abundance distribution, etc.) if they correspond to the same mean, variance and symmetry of a_{ij}

Predictions

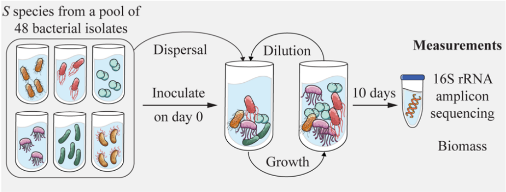
In particular, if you increase *number* and *variance* of interactions, you very predictably go from simple to complex dynamics



e.g. abundances fluctuating chaotically forever even in perfectly stable environment

Empirical test

Experimental setup: soil bacteria competition **(Hu et al, Science 2022)**

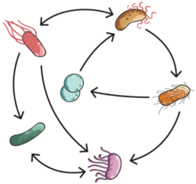


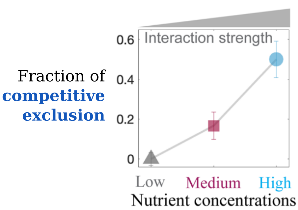
Jiliang Hu

Jeff Gore

Very well controlled environment & replicable results

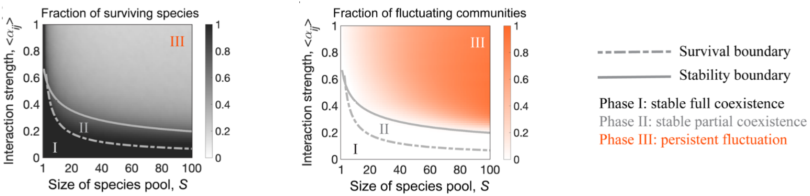
Unique feature: ability to **control how strongly species interact**



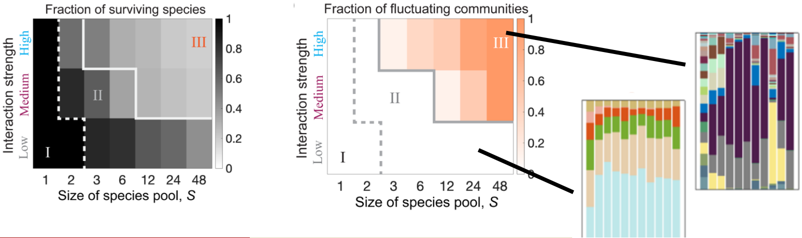


white = unique equilibrium, orange = fluctuations (in theory, chaos)

Random Lotka-Volterra Theory



Microbial experiments



Now that I gave this illustration, two questions:

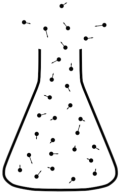
- how is that possible?
- how does that help us understand ecology?

I. How can complexity yield simplicity?

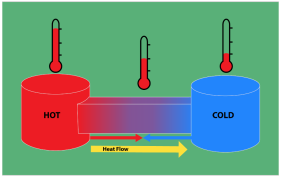
Idea originating from physics

When a system has *many* variables, a much simpler description is often possible

10^{23} variables:
position of every molecule

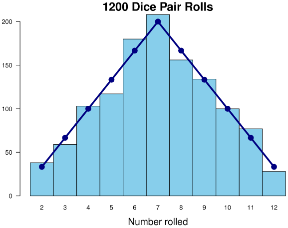


2 variables: temperature & pressure



uncountable factors, chaotic motion

1 probability distribution



Meaning of randomness

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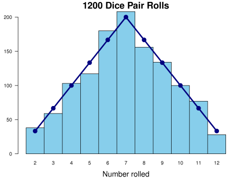
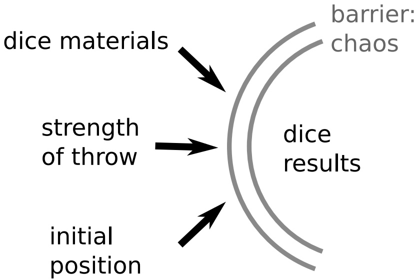


dice are simple *because* they are extremely sensitive to many details, making their movement chaotic and impossible to control

(vs e.g. basketball throws, where you need to know who is throwing, how far...)

Meaning of randomness



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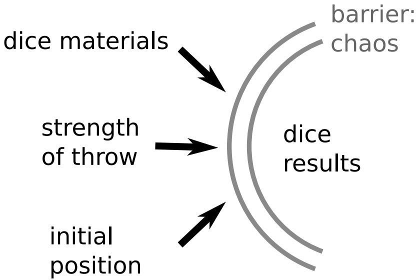

- “barrier” against details = chaos, motion unpredictable even if you know almost all details
- result = randomness, unpredictability becomes simplicity

“Random” means “too many factors”, so complex mechanistically that it becomes simple statistically

Random interactions

You already accept random effects from e.g. environment, question now: when does it make sense to think about random *species interactions*?

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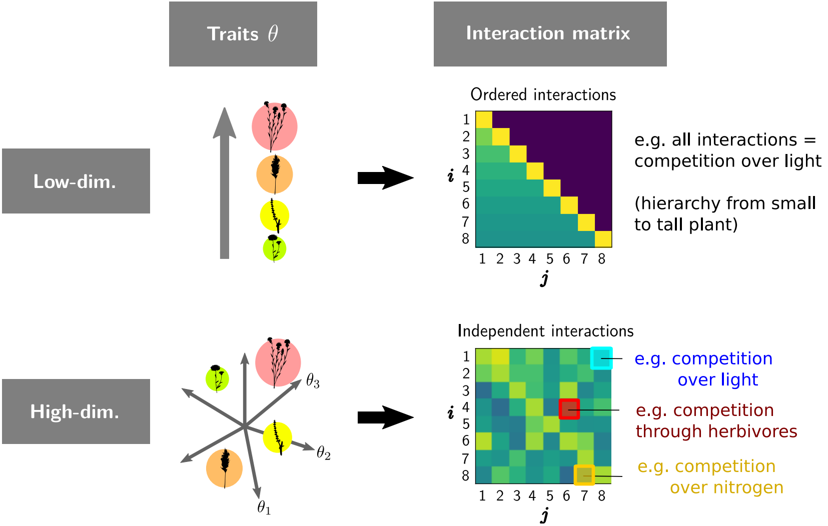
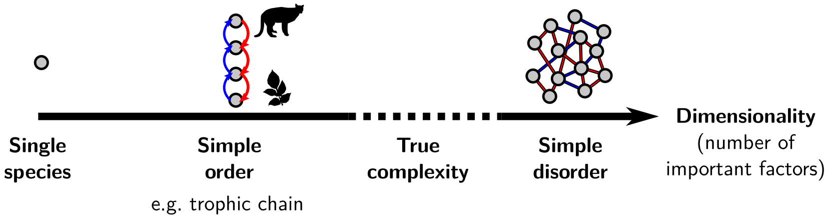


To conclude, modelling ecological interactions can start

- from "small simplicity" (e.g. a 3-species trophic chain)
- or from "large simplicity" = many-species "random-like" networks...



II. Many-species communities

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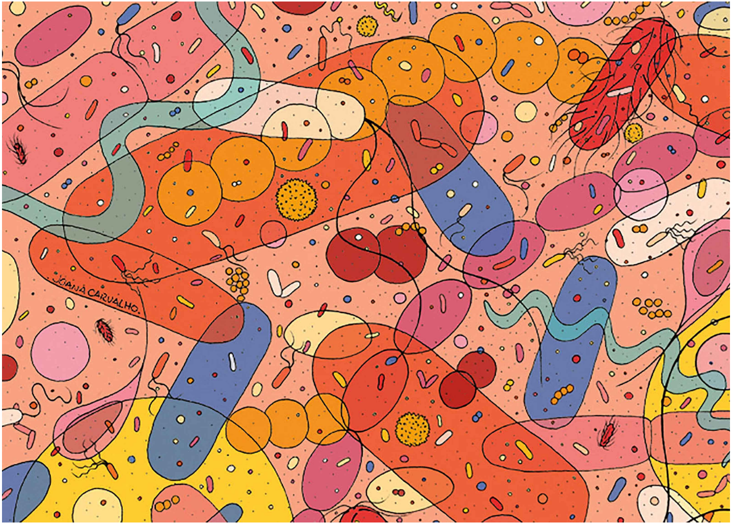
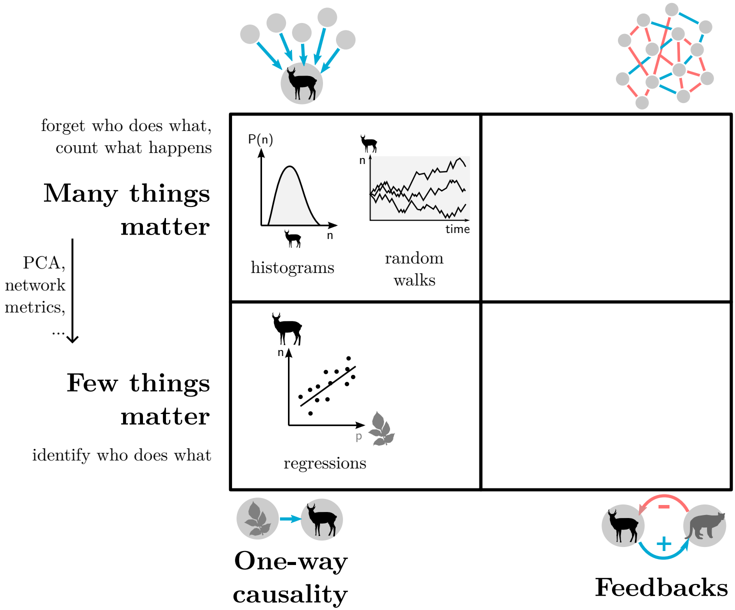
Part 1: What data are we trying to explain

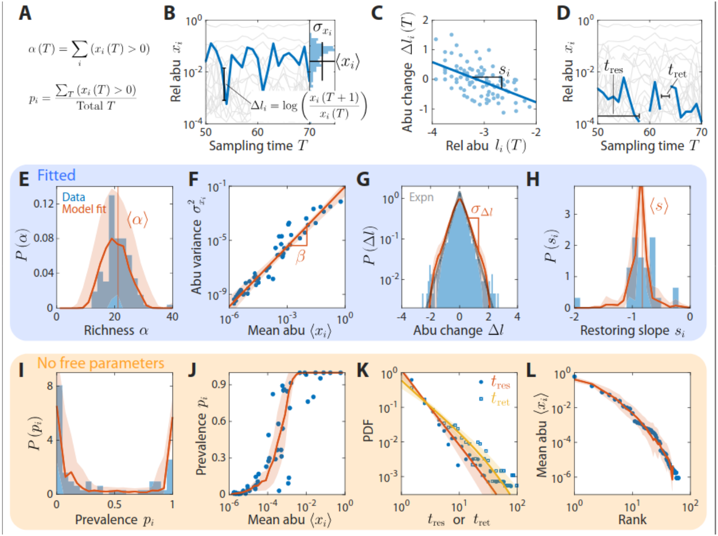
Forget about randomness for now, just study communities with many "species"



Lose focus on individual species, gain aggregate properties (top row)



Many patterns from histograms and aggregate stats (diversity, mean, variance...)



Ho P.Y. et al 2022 eLife

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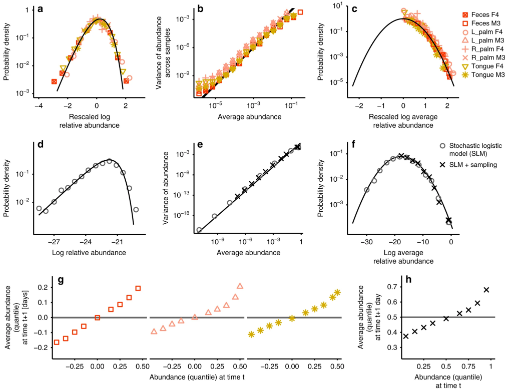
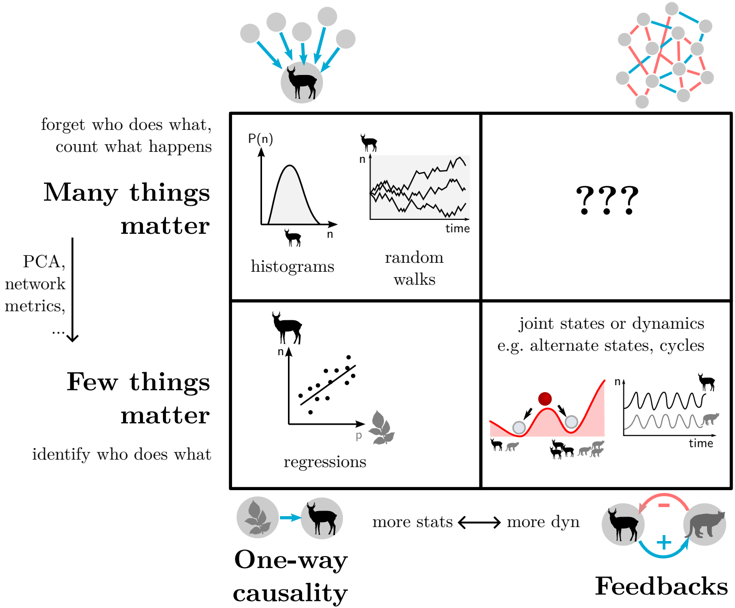
e.g. independent species with fluctuating growth rate reproduce many patterns

$$\frac{dN}{dt} = rN(1 - N/K) + N\xi(t)$$



Grilli 2020 Nat Comm – **model in black**

What is unique to interactions is the possibility of **emergent dynamics**



Dynamical properties

Can we predict how a whole ecosystem responds when we disturb it?

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  example: gut microbiome disturbed by sickness then re-colonized

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If these properties arise from the species being together, then interactions are what matters

II. Many-species communities

–

Part 2: How do we predict and explain such data

Model

Back to our Lotka-Volterra

$$\frac{dN_i}{dt} = r_i N_i (1 - a_{ii} N_i - \sum_{j \neq i}^{S} a_{ij} N_j) \tag{5}$$

Model

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$$\frac{dN_i}{dt} = 0 \Rightarrow \begin{cases} N_i = 0 & \text{extinct species} \\ N_i = (1 - \sum_{j \neq i}^{S} a_{ij} N_j)/a_{ii} & \text{surviving species} \end{cases} \tag{6}$$

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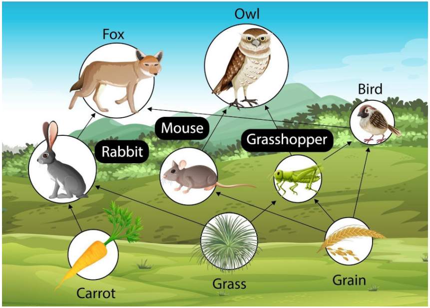
- maybe dynamical properties can be understood, e.g. how does $N_i(t)$ respond to perturbation, does it go to equilibrium or cycle...

Species interaction networks

How do we obtain the matrix of interactions a_{ij}?

- *Good news:* qualitative structure ($a_{ij} = 0$ or $\neq 0$) can be known for some interaction types, e.g. food webs



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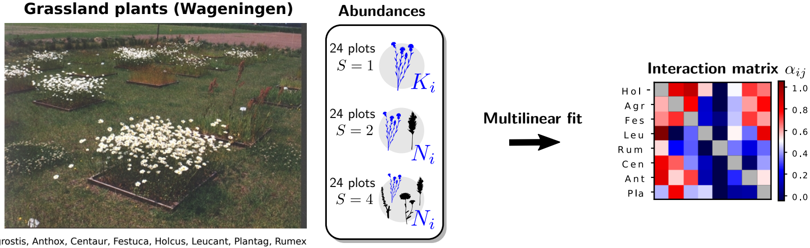
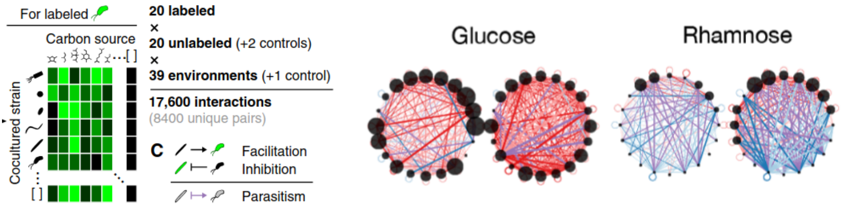
- *Bad news:* quantitative strength (a_{ij} values) is very rarely measured directly for every pair of species i, j (few experiments doing all that)



Agrostis, Anthox, Centaur, Festuca, Holcus, Leucant, Plantag, Rumex

How do we obtain the matrix of interactions a_{ij}?

- *Very bad news:* a_{ij} are **not context-independent parameters**

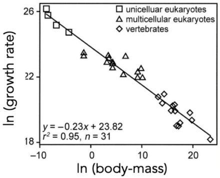


Kehe et al 2021 Sci Adv

Variance of 1 interaction across env. $\sim$ variance across all species in one env.

Species interaction networks

- Most of the time, assumptions are needed to put numbers into the model:
  - Metabolic scaling, r_i and a_{ij} given by body sizes of species i and j



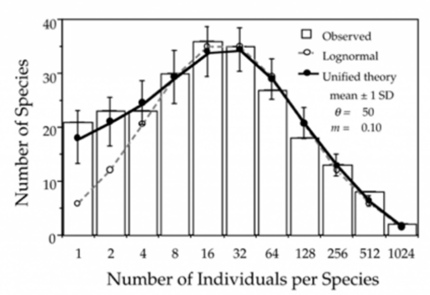
  - Ecopath model (see with Claire this afternoon)
  - ...

Neutrality

- Extreme simplification: neutrality, all species identical, e.g. $r_i = a_{ij} = 1$
- Different outcomes for different species only due to chance: random events of birth, death and migration

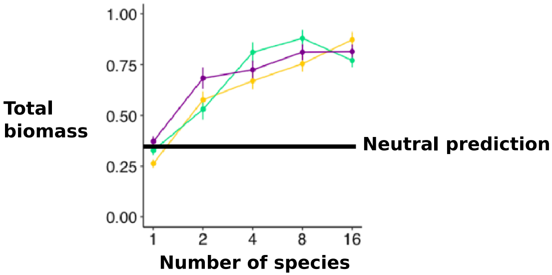
Neutrality

- Extreme simplification: neutrality, all species identical, e.g. $r_i = a_{ij} = 1$
- Different outcomes for different species only due to chance: random events of birth, death and migration
- Why use it? Because it can suffice to predict some patterns, e.g. abundance distributions



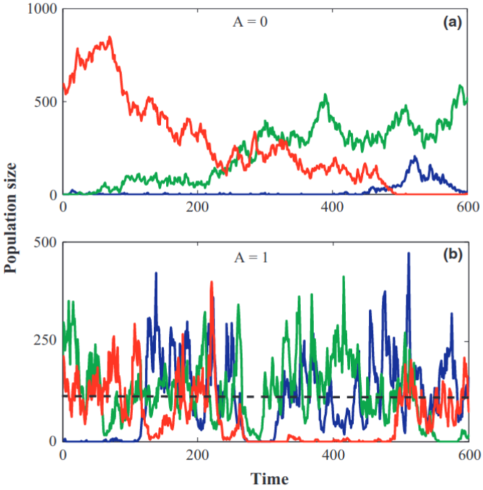
It fails but can be adapted for other patterns, e.g.

- More biomass when more species (original neutral theory = zero-sum game, total biomass is fixed)



It fails but can be adapted for other patterns, e.g.

- Temporal fluctuations from original neutral theory are too slow



Neutral theory prediction
(fixed environment, random birth-death)

More realistic timeseries
(fluctuating environment)

Neutrality

But neutrality will never give different *kinds* of dynamics (e.g. equilibrium or chaos)

Random interactions

Next simplest thing:

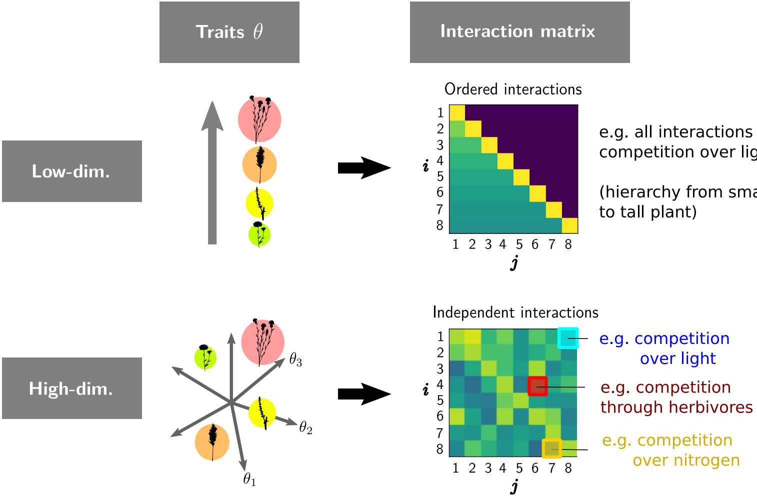
- while neutrality = identical interactions

$$a = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \tag{7}$$

- instead, take interactions a_{ij} that are different, but drawn at random

$$a = \begin{pmatrix} 0.29 & 0.54 & 0.53 & 0.02 & 0.40 \\ 0.57 & 0.86 & 0.90 & 0.81 & 0.76 \\ 0.53 & 0.11 & 0.42 & 0.44 & 0.09 \\ 0.15 & 0.72 & 0.84 & 0.27 & 0.94 \\ 0.87 & 0.85 & 0.61 & 0.36 & 0.63 \end{pmatrix} \tag{8}$$

Justification: interactions not “really” uncertain, but caused by many independent ecological traits, mechanisms, etc.



Predictions

As I said at the start, under broad conditions, results only depend on 4 parameters

- number of species S
- mean of interactions $\langle a_{ij} \rangle$ ($= \mathrm{E}[a_{ij}]$ if you prefer)
- standard deviation $\mathrm{std}(a_{ij})$
- and symmetry $\mathrm{corr}(a_{ij}, a_{ji})$

Illustration in static patterns

From a system of S equilibrium equations on all the abundances
(here $r_i = a_{ii} = 1$ for simplicity)

$$N_1 = 1 - \sum_{j \neq 1}^{S} a_{1j} N_j \tag{9}$$

$$N_2 = 1 - \sum_{j \neq 2}^{S} a_{2j} N_j \tag{10}$$

$$\ldots \tag{11}$$

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$$\ldots \tag{11}$$

We can go to a few equations on statistics of abundances, almost like this (a bit more complicated)

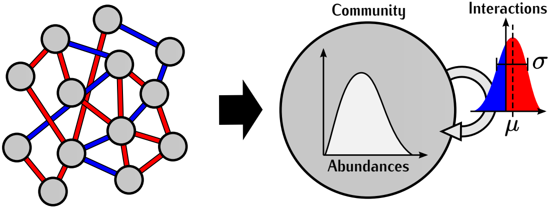
$$\langle N \rangle = 1 - S \, \langle a \rangle \langle N \rangle \tag{12}$$

$$\mathrm{var}(N) = S \, \mathrm{var}(a) \langle N^2 \rangle \tag{13}$$

where the **only parameters** are number of species S, and statistics of interactions $\langle a \rangle$, $\mathrm{var}(a)$ (or $\mathrm{std}(a)$)

Why?

Why does it work to just write equations on statistics (e.g. mean and variance) of abundance?

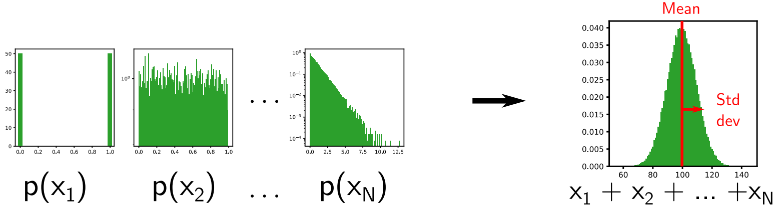


In a disordered network, all species are different but statistically equivalent

(no special role/position, any one at random is a fair sample from whole distribution)

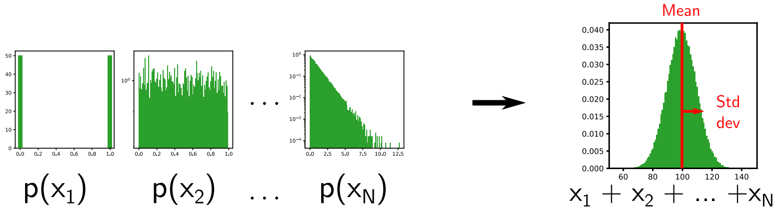
Why?

- Like Central Limit Theorem: many independent variables together create a Gaussian, with only 2 parameters: mean and variance



Why?

- Like Central Limit Theorem: many independent variables together create a Gaussian, with only 2 parameters: mean and variance

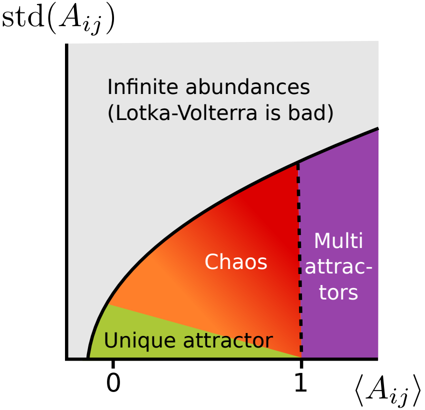


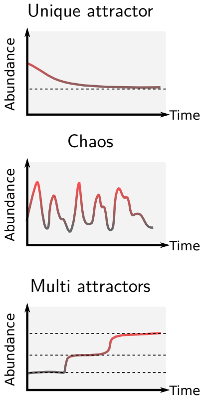
- Same is true with networks: many independent interactions together create a simple statistical result with only 3 parameters

How do we prove that result? Mathematical methods from physics

Predictions: dynamics

- Same reasoning works for dynamics, "$d\langle N\rangle/dt$, $d\text{var}(N)/dt$..."
- Hence, with few parameters, we can answer the question of how a many-species network behaves dynamically "by default" (when random)





NB: Chaotic phase shows "realistic" fluctuations

Random interactions

40 years of empirical successes for random interaction models:

- In physics: glass, large atoms
- In chemistry: large reaction networks
- In neuroscience: neural networks
- In computer science: complex satisfaction problems (NP-hardness)

Random interactions

40 years of empirical successes for random interaction models:

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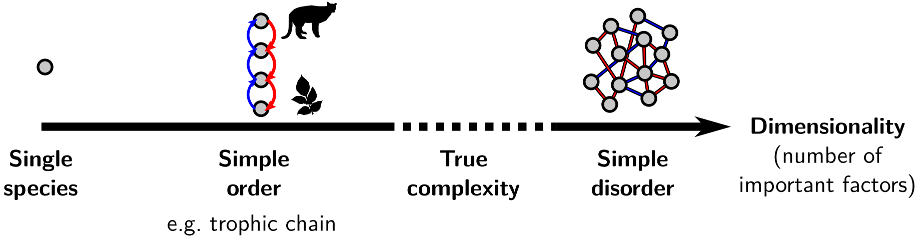
- In computer science: complex satisfaction problems (NP-hardness)

... but do we really believe that ecological systems can be modelled as completely random?

III. Order and disorder

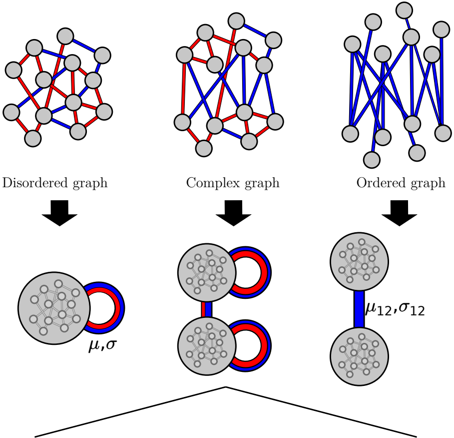
Combining order and disorder

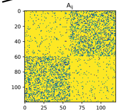
"There is a fundamental dichotomy between structure and randomness, which in turn leads to a decomposition of any object into a structured (low-complexity) component and a random (discorrelated) component."
– Terence Tao



Claim to test: Often, apparently complex systems behave like interpolation between simple order & disorder

Example 1: competitors and mutualists





Matrix μ

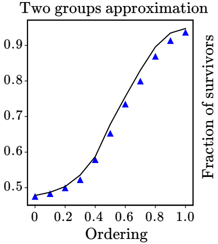
$$\begin{pmatrix} 0.4 & 2 \\ 2 & 0.4 \end{pmatrix}$$

Matrix σ

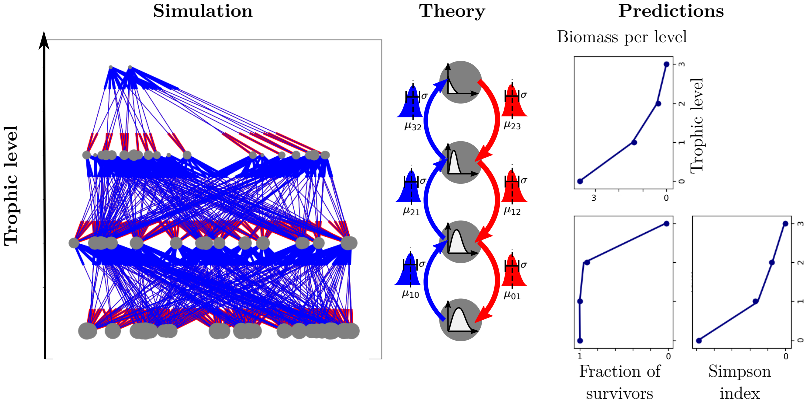
$$\begin{pmatrix} 0.1 & 0.4 \\ 0.4 & 0.1 \end{pmatrix}$$

Matrix γ

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

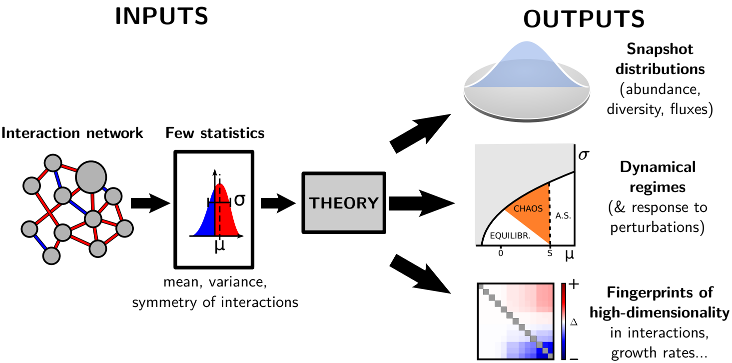


Example 2: food webs



(but also size hierarchy, nestedness, trade-offs...)

Random interactions = few fitting parameters, many testable outputs



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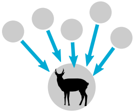
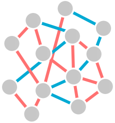




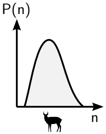
forget who does what,
count what happens

Many things matter

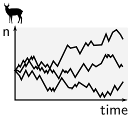
keystone spp/traits, PCA, network metrics, ...

Few things matter

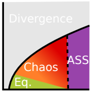
identify who does what



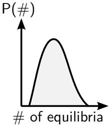
histograms



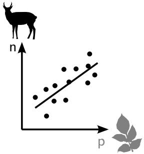
random walks



collective dynamical regimes

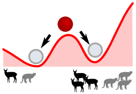


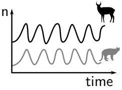
counting states

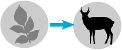


regressions

joint states or dynamics
e.g. alternate states, cycles

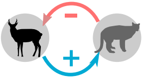






One-way causality

more stats ⟷ more dyn



Feedbacks

Take aways

- Observables in large systems:
  - snapshot patterns: distribution and statistics
  - dynamics: number & nature of attractors, sensitivity to perturbations
- Can we really identify ecological mechanisms from observables, or can we explain observables without knowing much about mechanisms?
  - For many observables, not all details of mechanisms matter; the art of modelling involves understanding when and which details are lost
- Randomness = particular case where we can prove that all details are lost except a few basic statistics
  - **drawing things at random does not mean you explore all possibilities!**
  - useful as null model; to know if network structure is important for a result, compare to result of random networks with similar statistics
  - can be mixed with simple structure (e.g. functional groups, nestedness...) to model "complex" networks
    $\Rightarrow$ what seems complex may be largely random